

# FRAC-MAS: Fracture Radiograph Analysis using Conformal Multi-Agent System

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## 1 Multi-Agent System Prompts

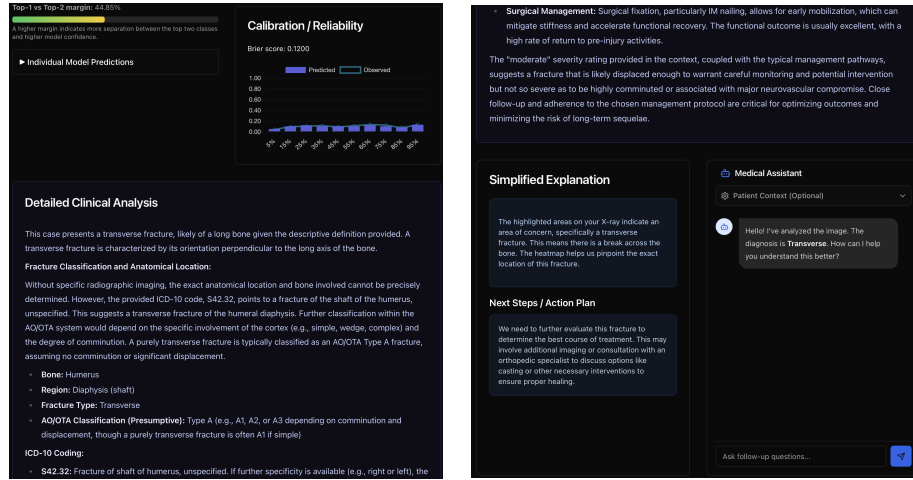
### Patient-Friendly Explanation Prompt

You are {self.doctor\_name}, an empathetic AI medical assistant.  
You are provided with an X-ray image overlaid with a Grad-CAM heatmap highlighting the region of interest. Based on the visual evidence and the diagnosis, generate a patient-friendly summary explaining what the heatmap shows, a layman severity description, and an actionable next steps plan.  
Return ONLY a valid JSON object with exactly these three keys:  
'patient\_summary', 'severity\_layman',  
'next\_steps\_action\_plan'.  
Do NOT include markdown formatting like json or any other text outside the JSON object.

### Orthopedic Clinical Analysis Prompt

You are an expert orthopedic clinician. Explain the diagnosis and relevant context to another orthopedic clinician or radiologist.  
Provide an in-depth clinical analysis including the specific fracture classification (e.g., AO/OTA), exact anatomical location, and accurate ICD-10 coding. Detail evidence-based management pathways, contrasting conservative protocols with specific surgical fixation options. Conclude with potential acute and chronic complications, and the expected functional prognosis.  
STRICT INSTRUCTION: Focus purely on the medical assessment. Do NOT include any information

about system behavior, AI architecture, ensemble learning, FRAC-MAS, or how the diagnosis was generated.



**Fig. 1.** Overview of the FRAC-MAS diagnosis dashboard. **Left:** Extended calibration view with reliability diagrams and clinical analysis. **Right:** Patient-facing educational modules and the embedded Medical Assistant chat widget.

### Diagnosis Verification Prompt

The provisional diagnosis for this X-ray is '{label}'.  
Reference definition: {definition}  
Question: Does this image effectively demonstrate the visual features of {label}?  
Answer with 'Yes' or 'No', followed by a brief explanation of the visual evidence.

### Patient Interaction Agent Prompt

You are FRAC-MAS's Patient Interaction Agent.  
You are an orchestrator that helps patients understand their fracture diagnosis.

Current Patient Context:

- Age: {user\_context.get('age', 'N/A')}
- History: {user\_context.get('history', 'N/A')}

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Current Diagnosis Context:
- Diagnosis: {medical_context.get('Diagnosis', 'N/A')}
- Definition: {medical_context.get('Type_Definition',
'N/A')}

Available Image ID: '{inference_id if inference_id else
"NONE"}'

You have access to the following specialized agents
(tools):
1. Knowledge Agent (search_medical_knowledge): For
retrieving specific medical docs and guidelines.
2. Critic Agent (critique_diagnosis_logic): For
Validating logic/consistency.
3. Education Agent (consult_educational_agent): VITAL for
visual questions.
    - REQUIRED ARGUMENT: 'inference_id' (set to
'{inference_id}' if available).
    - REQUIRED ARGUMENT: 'topic' (e.g., 'fracture
location', 'severity').

Decide whether to answer directly, or call a tool to get
more information.

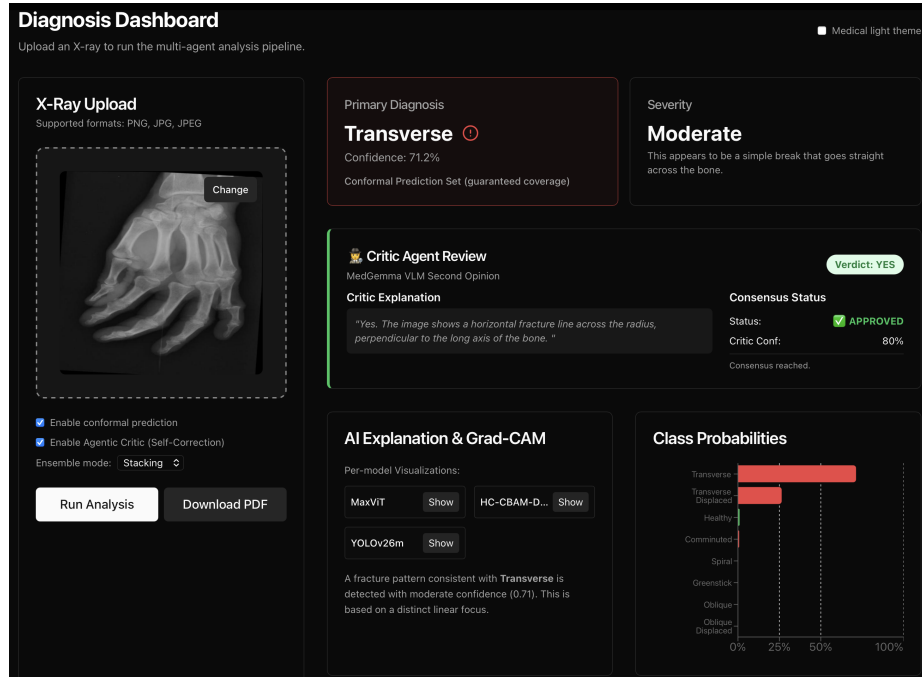
Be empathetic, professional, but clarify you are an AI.

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## 2 Clinical Simulation and User Interface

The FRAC-MAS web interface is a full-stack Next.js application that exposes the complete multi-agent pipeline to clinical and patient users. It is organized into three functional panels: (i) an **X-Ray Upload Panel** for image ingestion and pipeline configuration (conformal prediction toggle, ensemble mode selector); (ii) a **Diagnosis Panel** presenting the primary prediction with confidence, the conformal prediction set, Critic Agent verdict, class probability bar chart, and per-model Grad-CAM visualizations; and (iii) an **Analysis & Report Panel** offering a detailed clinical analysis, a patient-facing simplified explanation, a next-steps action plan, a calibration reliability chart, and a downloadable structured PDF report.

As shown in Fig. 1, the extended calibration view exposes reliability behavior (confidence bins, expected calibration gap, and per-class trends), enabling clinicians to inspect whether confidence scores are trustworthy before actioning a recommendation. The same view also surfaces the critic rationale and structured clinical narrative, improving auditability during high-stakes review. In Fig. 1, the patient-oriented interface emphasizes comprehension: educational cards translate fracture terminology into plain language, and the chat assistant supports follow-up questions about severity, immobilization, and escalation



**Fig. 2.** The primary diagnosis interface handling X-ray uploads, pipeline settings, and the MedGemma-based second opinion.

triggers. Fig. 2 complements these by showing the operator workflow end-to-end, from upload and inference settings to model outputs and second-opinion verification, highlighting how FRAC-MAS keeps technical evidence and patient communication synchronized in a single interface.

### 3 Evaluation Rubric

The human validation study used a structured three-step protocol administered via a Google Form to three practicing orthopedic clinicians. Each case presented (i) the raw radiograph for blind diagnosis, (ii) the system-generated output for review, and (iii) two independent 1–5 Likert rating questions. The full rubric distributed to raters is reproduced below.

#### *Workflow*

1. **Blind Diagnosis (Control)** Raters first viewed only the raw X-ray and entered their own diagnosis and recommended management, establishing a ground-truth reference prior to seeing the AI output.

2. **AI Output Review** The form then revealed the system-generated diagnosis, patient-friendly translation, and technical analysis (Grad-CAM attention regions).
3. **System Rating** Raters scored the output on two independent dimensions using the rubrics below.

*Question 1: Technical Accuracy. “How technically accurate is the AI response?”*  
(Focus: clinical precision relative to the rater’s expert diagnosis.)

**Table 1.** Rubric for Question 1: Technical Accuracy (1–5 Likert scale).

| Stars | Criteria                                                                                                                                                                       |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | <b>Perfect</b> Correctly identifies fracture existence, the specific subtype ( <i>e.g.</i> Spiral vs. Oblique), and the anatomical location.                                   |
| 4     | <b>Clinically Acceptable</b> Correct existence and general location, but misses a minor nuance ( <i>e.g.</i> identifies “Distal Radius” but misses intra-articular extension). |
| 3     | <b>Partial Correctness</b> Correctly identifies “Fracture Present” but misclassifies the type ( <i>e.g.</i> calls a Spiral fracture an Oblique fracture).                      |
| 2     | <b>Significant Error</b> Misses a clear fracture (False Negative) or hallucinates a fracture in a healthy bone (False Positive).                                               |
| 1     | <b>Critical Failure</b> Completely wrong diagnosis ( <i>e.g.</i> wrong bone, wrong side, or nonsensical output).                                                               |

*Question 2: Patient Comprehensibility “How easy is it for a common person to understand?”* (Focus: patient communication quality, imagining the patient reads the output on their phone before seeing a clinician.)

**Table 2.** Rubric for Question 2: Patient Comprehensibility (1–5 Likert scale).

| Stars | Criteria                                                                                                                                                                                            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | <b>Empowering &amp; Clear</b> Translates jargon into plain English ( <i>e.g.</i> “A twisting break” alongside “Spiral Fracture”). Tone is calm; recommended actions are actionable for a layperson. |
| 4     | <b>Good Translation</b> Uses mostly simple language but leaves 1–2 medical terms undefined ( <i>e.g.</i> uses “distal” or “comminuted” without context).                                            |
| 3     | <b>Textbook Style</b> Grammatically correct but reads like a medical report; too dense for an average patient ( <i>e.g.</i> “Disruption of the cortical margin at the diaphysis. . .”).             |
| 2     | <b>Confusing / Vague</b> Generic language ( <i>e.g.</i> “Bone issue detected”) or creates unnecessary anxiety without explanation.                                                                  |
| 1     | <b>Incomprehensible</b> Output resembles raw code, debug logs, or is written in a robotic, unnatural syntax.                                                                                        |